Foundation of Embodied AI: Course Map

2026-10-30

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Item	Details
Topic	Course Map
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Two learning tracks

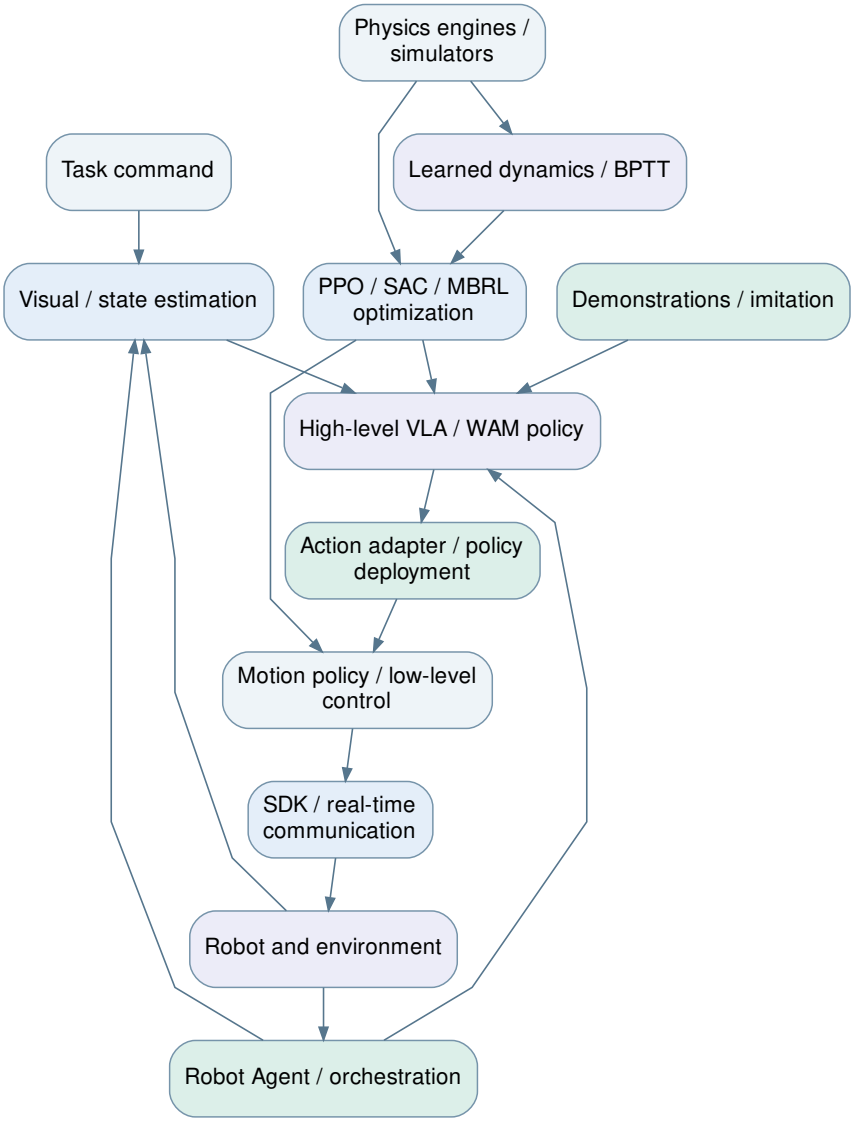
The course has two complementary parts that can be studied in parallel: **Theory & Algorithms (12 topics)** and **Engineering & Systems (8 topics)**. Choose a theory research, robotics engineering or full-stack path and use cross-references to fill prerequisites.

Courses are numbered 1–20 in the current display order, matching the README.

- **Part I: Theory & Algorithms (1–12)**: foundations, robot learning algorithms, embodied foundation models and post-training.
- **Part II: Engineering & Systems (13–20)**: tools, calibration, simulation training, real-time inference, communication and integration.

The collection remains 20 topics, 40 Markdown files and 40 PDFs. Both parts can be studied in parallel; choose a reading path by goal.

Embodied AI system map



- **Theory & Algorithms** explains vision, policy learning, models and control.
- **Engineering & Systems** implements simulation, sensing, compute, communication, deployment and robot feedback loops.
- **Policy training and real-robot feedback** connect both parts.

Action adapters and low-level interfaces depend on the robot and action space; VLA/WAM designs need not share a single motion-policy interface.

Recommended reading paths and connections

These optional paths cross both parts; they are not mandatory sequences.

Path A: Theory Research

Deep Learning Basics **1** → Imitation Learning **8** → Flow Matching **9** → PPO **5** / SAC **6** → Model-Based RL **7** → GR00T/SONIC **10** → World Action Models **11** → Post-training **12**.

Supplement Vision Foundation Models **2**, Robot Dynamics **3** and Low-Level Control **4** as the research question requires. Full mathematical derivations are advanced reading; practice need not wait until every derivation is complete.

Path B: Robotics Engineering

Git **13** / ROS 2 **14** as needed → Camera Calibration **15** → Robot Dynamics **3** / Low-Level Control **4** → Simulation & RL **16** → Motion Control RL **17** → Efficient Training & Inference **18** → Communication & Real-Time Systems **19** → Robot Agents **20**.

Focus on training environments, action semantics, actuator parameters, real-time frequencies, latency, synchronization and fault handling. PPO/SAC derivations belong to **5/6**; engineering topics **16/17** focus on tasks, training, evaluation and sim-to-real.

Path C: Full-Stack Embodied AI

Combine paths A/B through robot perception/control → simulation and training → imitation learning, RL and foundation models → post-training → real-time deployment and feedback. Connect both parts around the perception–policy–control–robot loop.

Theory–engineering connections

Theory topic	Engineering connection
Vision Foundation Models (2)	Camera Calibration (15), real-time inference (18)
Robot Dynamics (3) and Low-Level Control (4)	Simulation (16), motion-policy training (17), real-time systems (19)
PPO (5) / SAC (6) / Model-Based RL (7)	Simulation training (16), Motion Control RL (17)
Imitation Learning (8) / Flow Matching (9)	Policy-training workflows (16), efficient inference (18)
GR00T/SONIC (10) / World Action Models (11)	Real-time inference (18), system integration (19, 20)
Post-training (12)	Training/inference (18), real-robot feedback and communication (19)

These are knowledge connections, not claims that an engineering course implements every detail of a linked theory course.

Tutorial overview

Part I: Theory & Algorithms

1. Deep Learning Basics

Deep Learning Basics — Follow a robot task from linear models and MLPs through losses, backpropagation and reliable training. Study CNN/ViT visual tokens, Q/K/V and hand-worked Attention, Transformer blocks and multimodal action heads; connect pretraining, SFT/LoRA and RL post-training to deployment timing and controller interfaces. BPTT and differentiable world models form an advanced appendix.

2. Vision Foundation Models

Vision Foundation Models — Compare segmentation, pose estimation, stereo depth and 3D foundation models. Follow four robotics workflows spanning known objects, stereo geometry, unknown shapes and human demonstrations.

3. Robot Dynamics

Robot Dynamics — Build physical intuition from a grasp: actuator response, mass and inertia, joint coupling, Jacobian force mapping, contact and friction. Connect fixed- and floating-base dynamics to low-level control, simulator parameters and diagnosis of sim-to-real failures.

4. Low-Level Control

Low-Level Control — Connect motor servo loops, PD/PID and MIT Mode to impedance, admittance and contact control. Relate policy action semantics to actuator response, simulated hardware alignment and protected tuning.

5. PPO / Actor–Critic

PPO / Actor–Critic — Derive PPO-Clip and Actor–Critic learning from policy gradients, Bellman targets and GAE. Explain Gaussian policy updates, gradient isolation and asymmetric robot observations.

6. Q-learning / SAC

Q-learning / SAC — Trace Bellman equations, TD/Q-learning and fitted Q-iteration toward DQN and continuous-action actor–critic methods. Derive modern SAC's critic, actor and temperature losses; compare PPO, DQN and SAC; connect HIL-SERL and VLA/WAM action-chunk value learning, with mathematical details in appendices.

7. Model-Based RL

Model-Based RL — Study how learned dynamics support planning, synthetic training data and latent imagination through Dyna, PETS, MBPO and Dreamer. Derive BPTT through the feedback loop, then connect SHAC's short-horizon critic bootstrap to SuperMimic's learned surrogate dynamics. Distinguish model-fitting and policy-imagination horizons, and examine contact-gradient reliability and model bias.

8. Imitation Learning

Imitation Learning — Connect BC/DAgger and human corrections to ACT, Diffusion/Flow action chunks and multimodal VLA policies. Distinguish prediction/execution horizons, prefix continuity and stale conditioning; compare RTC, Training-Time RTC, VLASH/FutureRTC, RDP and RTR as complementary approaches to physical-time execution.

9. Flow Matching

Flow Matching — Derive conditional Flow Matching, velocity fields and the training objective. Explain how ODE integration transforms noise into conditional robot action sequences.

10. GR00T & SONIC

GR00T & SONIC — Use GR00T and SONIC to explain System 1/System 0 humanoid control. Connect high-level action chunks, motion representations, whole-body policies and low-level execution.

11. World Action Models

World Action Models — Study how future video information supervises robot actions, then compare joint, feature-conditioned and action-only inference. Use DreamZero and Dream-1 to discuss training and cross-embodiment transfer; distinguish Fast-WAM and the two Faster-WAM approaches, and compare VLA/WAM designs through data, robustness, compute, latency and contact-control requirements.

12. Post-training

Post-training — Compare SFT and RL for pretrained VLA and WAM models. Study six VLA RL routes, WAM-RL with online video SFT and Flow-SDE, then connect actor–critic semantics, RLinf integration and robot deployment limits.

Part II: Engineering & Systems

13. Git Basics

Git Basics — Learn Git's working tree, staging area and commit history, then follow a GitHub workflow covering branches, pull requests, conflict resolution and common recovery commands.

14. ROS 2 Basics

ROS 2 Basics — Understand nodes, topics, services, actions, QoS and TF. Connect sensors, state estimation, policy inference and controllers through ROS 2 interfaces.

15. Camera Calibration

Camera Calibration — Study camera intrinsics, distortion, extrinsics and hand–eye calibration. Relate RGB-D observations and coordinate transforms to robot-frame object poses.

16. Simulation & RL

Simulation & RL — Distinguish model descriptions, physics solvers, scenes and RL task frameworks across Isaac Sim/Lab, MuJoCo/mjlab, Genesis World and Newton. Follow robot modeling, observations/actions/rewards/resets, training and evaluation through sim-to-real interface alignment; compare tools by task requirements rather than a universal speed ranking.

17. Motion Control RL

Motion Control RL — Define closed-loop motion tasks through observations, goals, actions, rewards and resets. Distinguish from-scratch RL, Motion Tracking and Residual RL from PPO/SAC and model-based optimization; connect reference motions and generalist controllers to teacher–student learning, RMA and Sim-to-Real. Use G1 velocity tracking and TRON2/Revo3 screwdriver manipulation to organize evaluation of success, generalization and recovery.

18. Efficient Training & Inference

Efficient Training & Inference — Connect Transformer and VLA/WAM training costs, FlashAttention and valid caching to DreamZero acceleration, few-step inference, asynchronous robot control, RTC and cloud latency.

19. Communication & Real-Time Systems

Communication & Real-Time Systems — Separate ROS 2/DDS, SDKs and CAN/EtherCAT buses. Explain timestamps, data freshness, latency and jitter, multirate scheduling, asynchronous inference and fault handling in robot feedback loops.

20. Robot Agents

Robot Agents — Distinguish foundation models, coding agents and agent harnesses. Connect RAX exploration, ASPIRE trace repair and RPent toolkits, memory, Flash Mode and the RLinf data flywheel to reusable robot workflows.